

9 [page Replacement Policy]

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4,5 (No of Frames 3).

CODE:

```
GNU nano 8.7
#include <stdio.h>

int main() {
    int pages[50], frames[10];
    int n, framesCount = 3;
    int i, j, k, pos, faults = 0, hits = 0;
    int flag1, flag2;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    for(i = 0; i < framesCount; i++)
        frames[i] = -1;

    for(i = 0; i < n; i++) {
        flag1 = flag2 = 0;

        for(j = 0; j < framesCount; j++) {
            if(frames[j] == pages[i]) {
                hits++;
                flag1 = flag2 = 1;
                break;
            }
        }

        if(flag1 == 0) {
            for(j = 0; j < framesCount; j++) {
                if(frames[j] == -1) {
                    faults++;
                    frames[j] = pages[i];
                    flag2 = 1;
                    break;
                }
            }
        }

        if(flag2 == 0) {
            int least[10];
            for(j = 0; j < framesCount; j++) {
```

```

GNU nano 8.7

    if(flag2 == 0) {
        int least[10];
        for(j = 0; j < framesCount; j++) {
            least[j] = -1;
            for(k = i - 1; k >= 0; k--) {
                if(frames[j] == pages[k]) {
                    least[j] = k;
                    break;
                }
            }
        }

        pos = 0;
        for(j = 1; j < framesCount; j++)
            if(least[j] < least[pos])
                pos = j;

        frames[pos] = pages[i];
        faults++;
    }

    printf("Frame status: ");
    for(j = 0; j < framesCount; j++)
        printf("%d ", frames[j]);
    printf("\n");
}

float hitPercent = ((float)hits / n) * 100;
float faultPercent = ((float)faults / n) * 100;

printf("\nTotal Page Hits = %d\n", hits);
printf("Total Page Faults = %d\n", faults);
printf("Hit Percentage = %.2f%%\n", hitPercent);
printf("Fault Percentage = %.2f%%\n", faultPercent);

return 0;
}

```

OUTPUT:

```

rachn@LAPTOP-6NDD8TJ3 MSYS ~
$ cd /c/Users/Rachna/Desktop
-bash: cd: /c/Users/Rachna/Desktop:

rachn@LAPTOP-6NDD8TJ3 MSYS ~
$ nano optimal.c

rachn@LAPTOP-6NDD8TJ3 MSYS ~
$ ^C

rachn@LAPTOP-6NDD8TJ3 MSYS ~
$ gcc optimal.c -o optimal

rachn@LAPTOP-6NDD8TJ3 MSYS ~
$ ./optimal
Enter number of pages: 7
Enter page reference string:
7 0 1 2 0 3 0
Frame status: 7 -1 -1
Frame status: 7 0 -1
Frame status: 7 0 1
Frame status: 2 0 1
Frame status: 2 0 1
Frame status: 2 0 3
Frame status: 2 0 3

Total Page Hits = 2
Total Page Faults = 5
Hit Percentage = 28.57%
Fault Percentage = 71.43%

rachn@LAPTOP-6NDD8TJ3 MSYS ~
$ |

```